

2.3.2 角动量守恒定律

Conservation of Angular Momentum

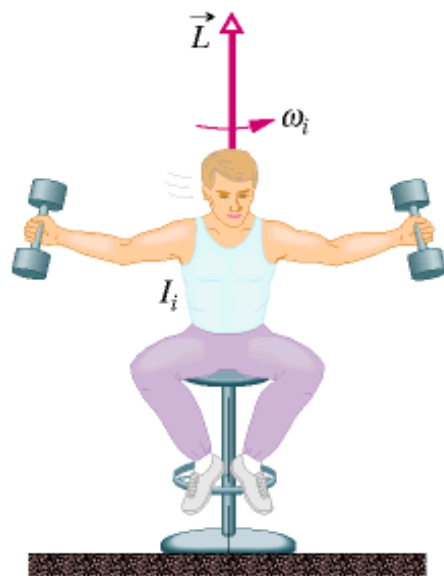
设质点系沿定轴 z 轴转动

$$\int_{t_1}^{t_2} M_z dt = L_{2z} - L_{1z}$$

若 $M_z=0$, 则 $L_{2z} = L_{1z}$

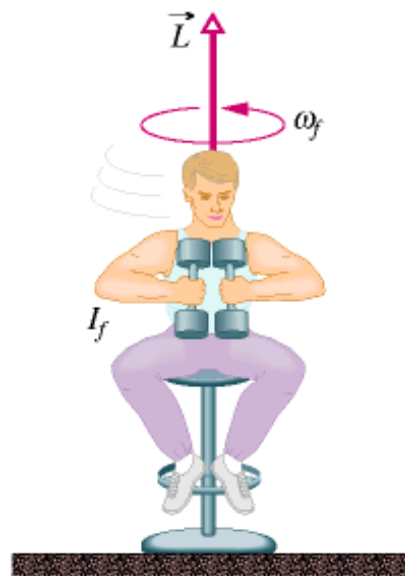
若质点系所受的对某一轴的合外力矩为零,
则它对这一固定轴的角动量保持不变。

例如：茹可夫斯基椅

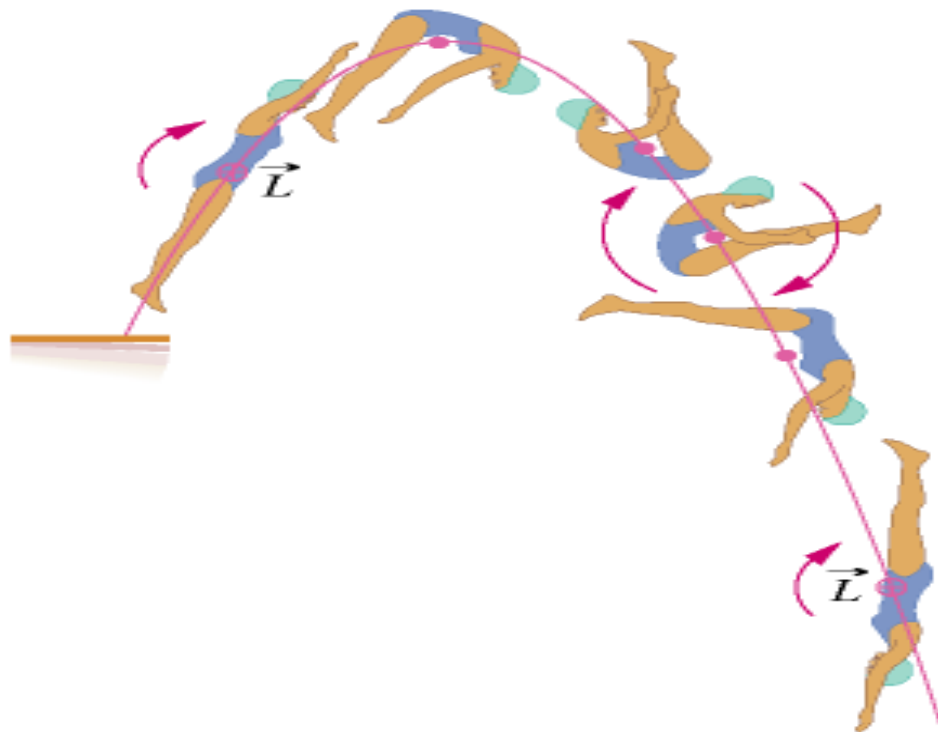


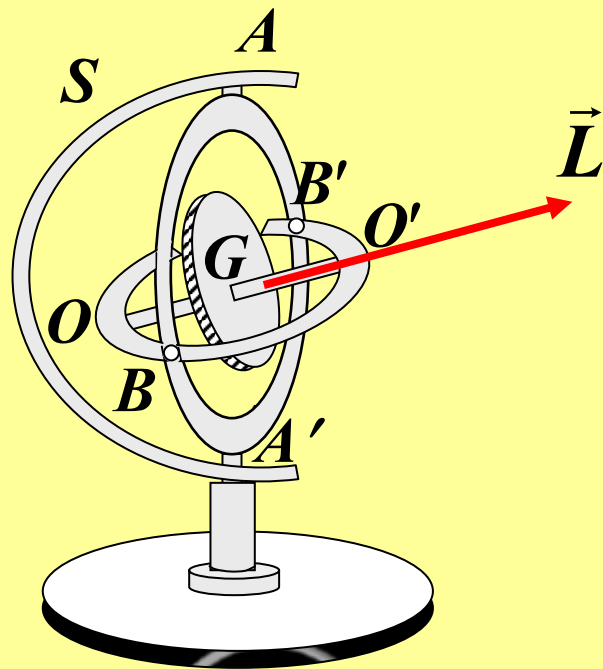
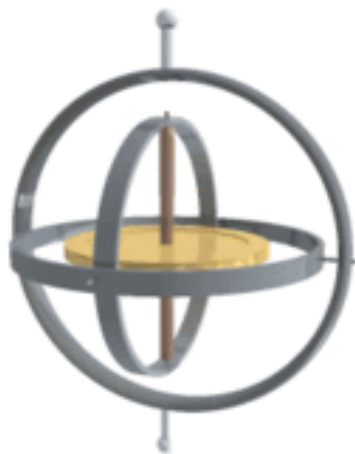
Rotation axis

(a)



(b)





常平架回转仪

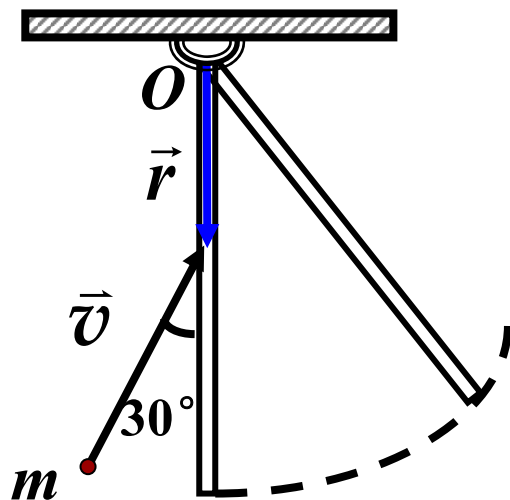


例：一匀质细杆可绕 O 点在竖直面内转动，其质量为 M ，长度为 l ，起始时静止于垂直位置。现有一质量为 m 的泥团以速度 v 与竖直方向成 30° 角方向击中细杆中央并粘住，求：细杆开始摆动时的角速度。

解：碰撞前后杆+泥团系统的角动量守恒

初态：细杆角动量为零

$$\vec{L}_o \begin{cases} \text{大小: } L_o = r p \sin \alpha = \frac{l}{2} m v \sin 150^\circ \\ \text{方向: } \odot \end{cases}$$

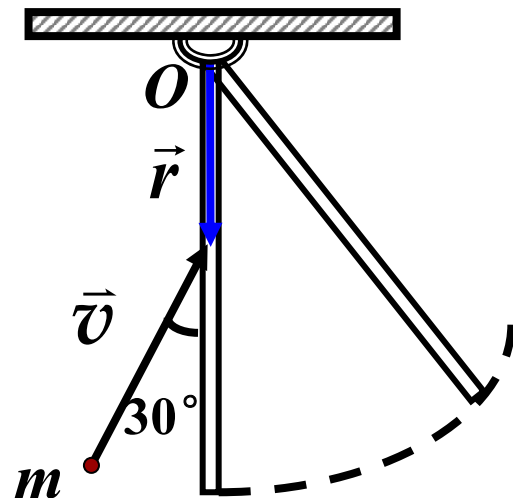


末态：泥团： $\vec{L}_m$ $\left\{ \begin{array}{l} \text{方向：} L_m = |\vec{r} \times m \vec{v}'| \\ \text{大小：} \odot \end{array} \right.$ $r = \frac{l}{2}, v' = \frac{l}{2} \omega$

$$L_m = \frac{l}{2} \omega \frac{l}{2} m \sin 90^\circ = \frac{l^2}{4} m \omega$$

杆的角动量：

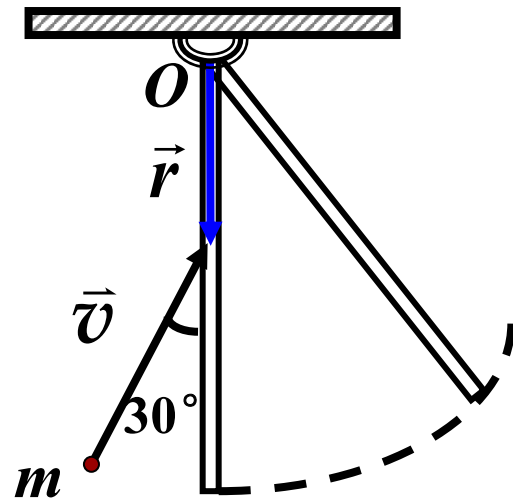
$\vec{L}_M$ $\left\{ \begin{array}{l} \text{大小：} L_M = J\omega = \frac{1}{3} M l^2 \omega \\ \text{方向：} \odot \end{array} \right.$



角动量守恒 $L_O = L_m + L_M$

$$\frac{l}{2} m v \sin 150^\circ = \frac{l^2}{4} m \omega + \frac{1}{3} M l^2 \omega$$

$$\omega = \frac{3 m v}{(3 m + 4 M) l}$$

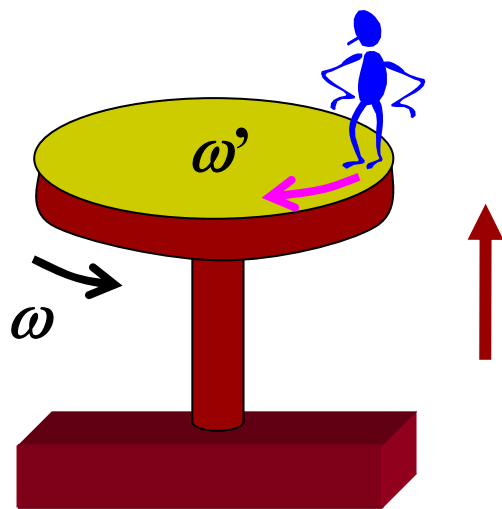


例：转台（ M, R ），人（ m ），轴的阻力不计，若人沿转台跑一圈，问：相对地面来说，人和转台各转了多少度？

解：人与转台组成的系统角动量守恒

$$0 = J\omega - J'\omega' \quad J = \frac{1}{2}MR^2, \quad J' = mR^2$$

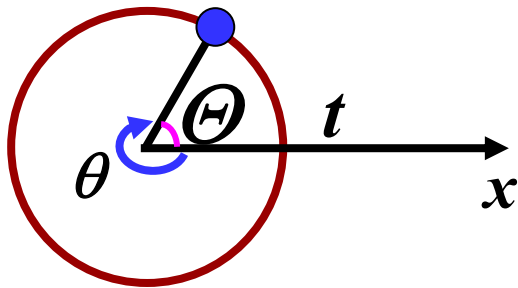
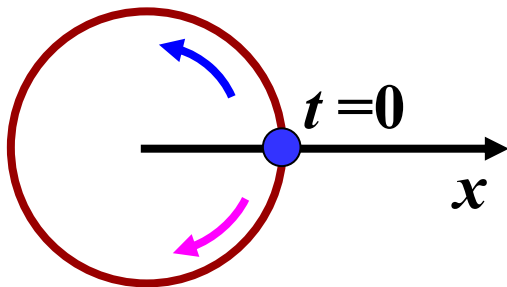
$$\frac{\omega}{\omega'} = \frac{J'}{J} = \frac{2m}{M} \quad \omega = \frac{2m}{M}\omega'$$



$$\int_0^T \omega dt = \frac{2m}{M} \int_0^T \omega' dt$$

$$\begin{cases} \Theta = \frac{2m}{M} \theta \\ \Theta = 2\pi - \theta \end{cases}$$

$$\begin{cases} \theta = \frac{2\pi M}{M + 2m} \\ \Theta = \frac{4\pi m}{M + 2m} \end{cases}$$



$$\omega = \frac{2m}{M} \omega'$$

